Q1.

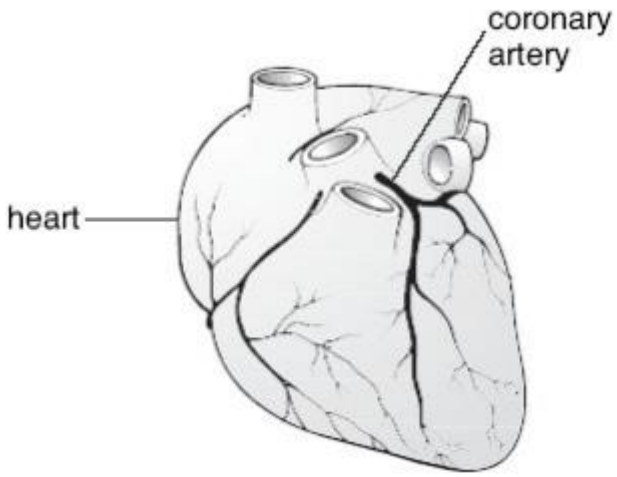
(a) Carbon monoxide, nicotine and tar get into the lungs when a person smokes.

Draw a line from each substance to the effect of the substance on the body.
Draw only **three** lines.

substance	effect of the substance
carbon monoxide	causes addiction to smoking
nicotine	causes influenza (flu)
tar	causes lung cancer
	causes red blood cells to carry less oxygen

3 marks

(b) The coronary arteries carry blood to the heart muscle.
The drawing below shows the heart and coronary arteries.



(i) Diagram 1 shows a section through a coronary artery.

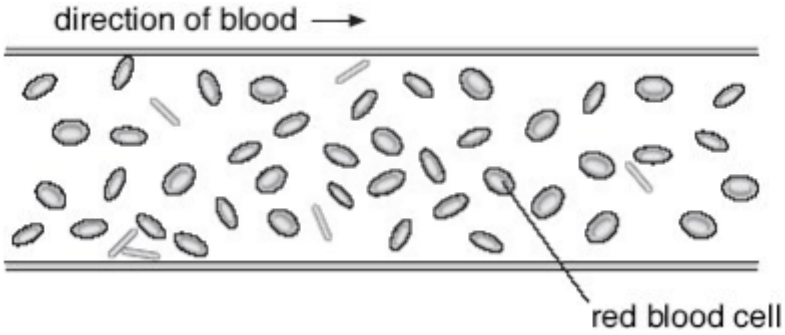


diagram 1

Smoking can cause damage to the coronary artery.
Diagram 2 shows a section through part of a damaged artery.

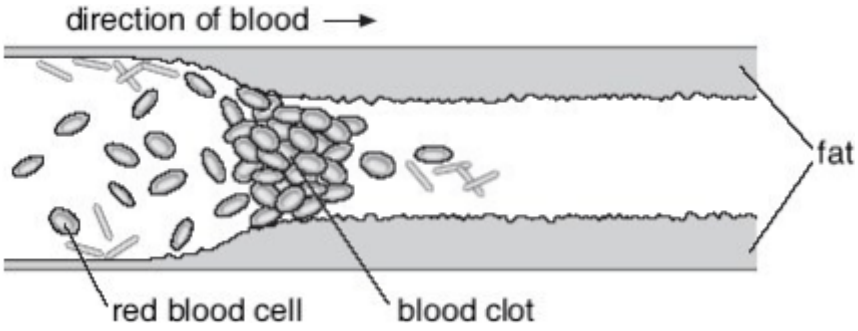


diagram 2

not to scale

Look at diagram 2. A blood clot has formed.

Give **one** other change in the coronary artery.

.....

.....

1 mark

(ii) Respiration takes place in the muscle cells of the heart.

Explain why a blood clot in the coronary artery prevents these cells respiring normally.

.....

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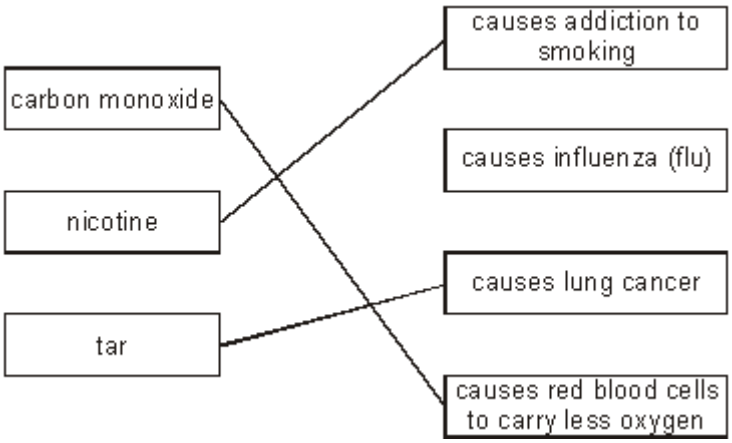
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2 marks
maximum 6 marks

Q1.

(a)



if more than one line is drawn from any substance, award no mark for that substance

3 (L5)

(b) (i) any **one** from

- fat has been deposited
  accept 'fat'
- the artery has become narrower
  accept 'it is blocked'

1 (L5)

(ii) any **two** from

- less oxygen to the cells or heart muscle
  accept 'no oxygen'
  *do **not** accept 'less air to the cells'*
- less glucose to the cells **or** heart muscle
  accept 'no glucose'
- less blood to the cells **or** heart muscle
  accept 'it stops the circulation of blood'
  accept 'blood cannot flow through'
  accept 'it stops blood getting to the heart'
  'nothing can get through' is insufficient
  accept 'there is a build-up of carbon dioxide'
  accept for two marks 'they get less oxygenated blood'

2 (L6)

[6]